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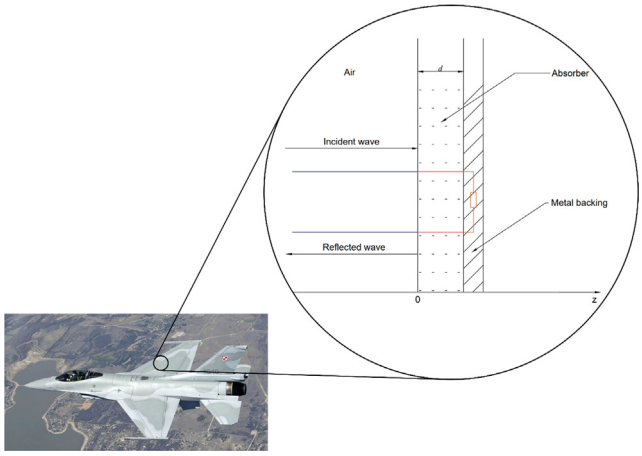
Feature Article

Deep understanding of impedance matching and quarter wavelength theory in electromagnetic wave absorption

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GRAPHICAL ABSTRACT



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ABSTRACT

It is of great significance for absorber designers to correctly understand impedance matching and its related calculation methods, because excellent impedance matching effect is an important prerequisite for absorber to consume electromagnetic wave (EMW). However, there are some problems in literatures, such as the lack of deep understanding of impedance matching and application conditions of Eq. (3) for the calculation of reflection loss (RL). Based on the transmission line theory of electronic circuit, this paper expounds the definition, symbol, property of various impedances and their influence on impedance matching effect. On the basis of simplifying the actual situation in the process of EMW absorption to the correct transmission line model, the original transmission line model of RL calculation formula and its limitation are pointed out. The doubts about the validity of the quarter wavelength theory in literatures are explained from three aspects: the reduction of EMW length, the multiple reflection and the impedance transformation of absorber.

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1. Introduction

Impedance matching is one of the key factors in the design of absorbers [1,2], but there are still some misunderstandings in some papers. The wrong understanding of the functions of various impedances will not only lead to the wrong use of the symbols in the calculation formula, but also bring the understanding error to the readers. As shown in Fig. 1, the EMW absorption model with metal backing is often used to calculate and evaluate the performance of absorbers [3,4]. However, the corresponding relationship between the circuit elements in the theoretical model of transmission line and the actual situation of air, absorbers and metal backing has not been mastered by everyone, and the input impedance, characteristic impedance and load impedance are confused. Some people believe that the absorber acts as a load in the transmission line model and uses Z_L to represent its characteristic impedance [5]. This idea apparently ignores the role of the metal backing and mistakenly assumes that the metal backing does not play a role in the RL calculation. The idea that the absorber acts as a load in the transmission line model and Z_L is used to represent its characteristic impedance appeared in some papers. In this case, the role of the metal backing is obviously ignored, and it is wrongly believed that the metal backing does not play a role in the calculation of RL. Some people think that the input impedance Z_{in} is equivalent to the characteristic impedance Z_M of the absorber. When $Z_{in} = Z_0$, the impedance matching without reflection is realized. The relationship between the input impedance and the characteristic impedance is not fully understood in this idea.

Starting from the transmission line theory of electronic circuit [6] this paper combines all kinds of media in the actual working environment of EMW absorption with the transmission line model, and clarifies the corresponding relationship among them [7]. For the single-layer absorber structure shown in Fig. 1, both the air and the absorber attached to the metal backing act as the transmission lines, and the metal backing acts as the load. When the normalized input impedance of the absorber and the metal backing is equal to the characteristic impedance of the air, the impedance matching and no reflection can be realized. It is also based on this understanding that the widely used RL calculation Eq. (2) and Eq. (3) obtained by Steffensen acceleration method are available [8,9]. Moreover, this calculation formula can only be used to characterize the comprehensive EMW absorption performance of the absorber in the model shown in Fig. 1, rather than the performance of the absorber itself, which has certain limitations.

In the design of absorber, quarter wavelength matching layer often acts as electromagnetic interference and impedance transformation to eliminate reflection, but it is also considered that quarter wavelength theory has defects. Because in vacuum, most of the quarter wavelength of EMW in the frequency range of 2–18 GHz is distributed in the centimeter scale. At present, the thickness of absorber is generally in millimeter scale, which is difficult to reach the theoretical thickness. There is no doubt that the quarter wavelength theory is correct, and the controversy can be explained by multiple reflections, wavelength changes and impedance transformation [6].

2. Results and discussion

2.1. Impedance matching

The premise of correctly understanding impedance matching is to understand the definition of input impedance Z_{in}, load impedance Z_L, characteristic impedance (characteristic impedance Z_0 of air and characteristic impedance Z_M of absorber) and the corresponding relationship between various impedances and transmission line model. In the simplified transmission line model of EMW absorption (Fig. 1), the load impedance and characteristic impedance are related to the inherent properties of the material [10] such as electromagnetic parameters [11] but not to the thickness of the absorber. The influence of absorber thickness on impedance matching is achieved by affecting the input impedance. When the medium of EMW transmission channel changes, the input impedance can replace the common characteristics of all different media along the propagation direction of EMW with Z_{in} equivalent. It can also be said that the input impedance has the same effect on the medium on the propagation path of EMW, so it is very convenient to analyze and calculate the impedance matching effect. Whether it is a single-layer or a multi-layer absorber, whether there is a metal back plate in the EMW absorption model, the impedance matching effect can be obtained by calculating the input impedance at the interface between the absorber and the air [12].

Due to the difference in the structure of the absorber and the actual working environment of the absorber, different transmission line models will inevitably be simplified. Obviously, there will be different RL calculation methods to characterize the absorber's absorbing performance. The input impedance of the starting position of the absorber must be obtained first when calculating RL. However, when the input impedance is used instead of the impedance of the subsequent EMW transmission medium, it must be carried out on the premise that the transmission path of EMW is known and all the media on the transmission path must be considered. In other words, when calculating the input impedance of the transmission line mode of the lossy absorber, it is assumed that the transmission end point of the EMW is known. Taking Fig. 2a (the color line in the picture represents the transmission line) as an example, when the absorber is a high loss absorbing material, the EMW is difficult to pass through the absorber, and the input impedance is the characteristic impedance of the absorber. When

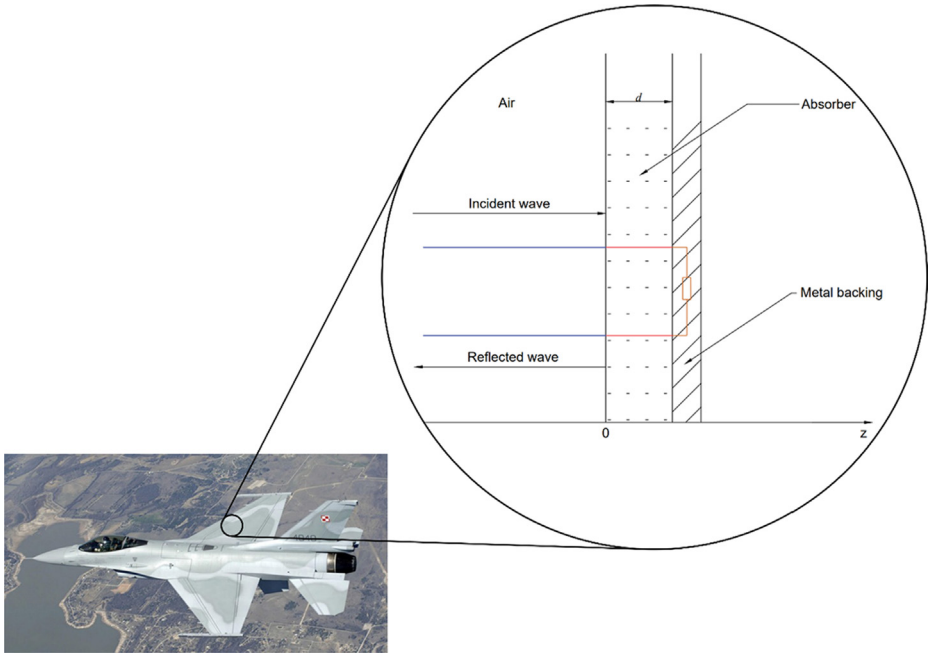


Fig. 1. Transmission line model with metal backing. Reprinted from [10] Copyright (2020), with permission from Elsvier.

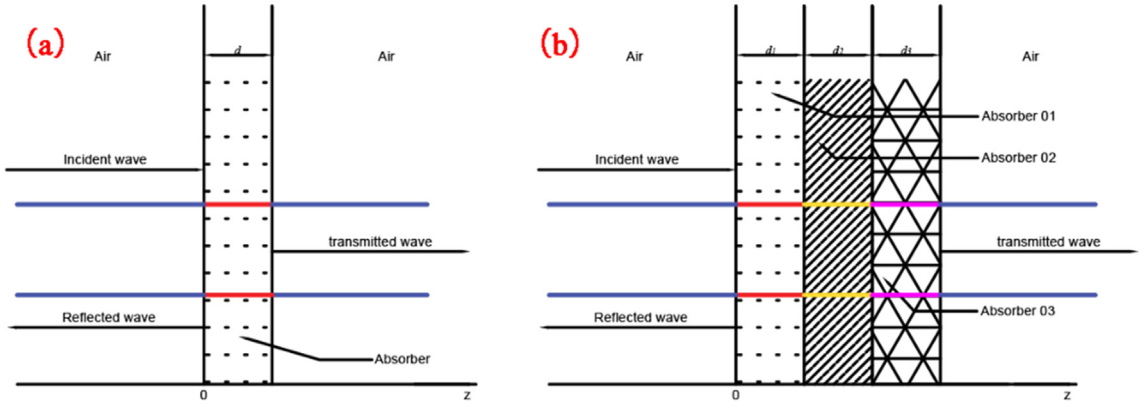


Fig. 2. (a) Single layer absorber transmission line model without metal backing and (b) multilayer absorber transmission line model without metal backing.

the absorber is a low-loss absorbing material, EMW will penetrates the absorber and enter the free space. At this time, the characteristic impedance of the material and the impedance of the free space need to be considered when calculating the input impedance. What's more, if there may be other media in the free space after the EMW passes through the absorber, its influence should also be taken into account. Similarly, it is difficult to determine whether the EMW can penetrate the multilayer material for the multilayer absorber shown in Fig. 2b. Because it is difficult to know whether the EMW will penetrate the material before the test, it is necessary to assume that the transmission end point of the EMW is known before the next calculation. On the one hand, the premise of this assumption may lead to inaccurate calculation results, on the other hand, the characterization of microwave absorption performance will not be unified. Therefore, in order to facilitate calculation and characterization, the transmission line model with absorber attached on the metal backing is often used.

For single-layer homogeneous absorbers, the widely used transmission line model is shown in Fig. 1. However, there are still some people who misunderstand the model. For example, some people do not know the function of the metal backing, mistakenly think that the absorber is the load, so that they use the wrong symbol in the calculation process, and even make calculation errors. In Fig. 1, the metal backing can be regarded as an ideal conductor with infinite conductivity. According to the skin effect, the penetration depth of EMW is basically equal to 0, and the EMW will be completely reflected at the interface between the absorber and the metal backing. According to the transmission line theory of electronic circuit, the metal backing can be regarded as a short-circuit load, and the load impedance $Z_L = 0$. EMW only propagates in air and absorber, both of which act as transmission lines. The characteristic impedance of air is Z_0 and that of absorber is Z_M. Different from the ideal circuit, the EMW will be consumed when it propagates in the air and absorber. Therefore, the model is equivalent to the lossy transmission line circuit with the change of transmission medium, and the propagation constant should be used to correct the calculation. In this model, on the one hand, the metal backing simulates the possible transmission environment of EMW in actual use, on the other hand, it simplifies the calculation of input impedance. For a single-layer absorber, the use of a transmission line model with a metal backing makes the transmission path of EMW clear, because EMW will inevitably be

reflected when they encounter the metal backing, and there is no need to consider the medium behind the metal backing. It is also conducive to the unity of performance characterization. The input impedance at the interface between air and absorber is calculated:

$$Z_{in} = Z_M \frac{Z_L - jZ_M \tan(kz)}{Z_M - jZ_L \tan(kz)} \xrightarrow{Z_L=0} Z_{in} = -jZ_M \tan(kd) \quad (1)$$

In most papers, Eq. (2) and Eq. (3) are used to calculate the RL of absorbers [13–18]. It should be noted that Eq. (2) is based on the EMW absorption model shown in Fig. 1, and Eq. (3) is the numerical solution obtained by Steffensen acceleration method on the basis of Eq. (2). Therefore, the two equations can only be used to calculate the RL of the absorber attached to the metal in the actual situation, rather than the general calculation formula. The RL here is only used to characterize the EMW absorption performance of the absorber under the combined action of the absorber and the metal backing in Fig. 1, but cannot be generalized in all cases. For example, Eq. (2) and Eq. (3) do not hold when the absorber changes from single layer to multi-layer (Fig. 2b). For the different application conditions of the absorber in the actual situation or for the absorber of different structure, the transmission line model should be first simplified, and then start from the original calculation formula of input impedance to get different RL calculation formulas.

$$Z_{in} = Z_0 \sqrt{\frac{\mu_r}{\varepsilon_r}} \tanh\left(j\frac{2\pi fd}{c}\sqrt{\mu_r \varepsilon_r}\right) \quad (2)$$

$$RL = 20\log|(Z_{in} - Z_0)/(Z_{in} + Z_0)| \quad (3)$$

2.2. Quarter wavelength theory

Some people doubt the validity of the quarter wavelength theory. The main reason is that most of the quarter wavelength of EMW in the frequency range of 2–18 GHz in vacuum is distributed in the centimeter scale. However, the thickness of wave absorbers currently used is generally in the millimeter scale, which is difficult to reach the theoretical thickness. There is no doubt that the quarter wavelength theory is correct and can be explained from the following aspects.

Firstly, the propagation speed of EMW in the absorber is less than that in the vacuum. In this case, the thickness of the quarter wavelength matching layer will be reduced, which will meet the interference cancellation of partial frequency EMW. Secondly, EMW will be reflected multiple times at the two interfaces between the absorber and air and the absorber and the metal backing. Therefore, the total distance of the electromagnetic wave from the absorber to the air after multiple reflections may reach half of the wavelength. Finally, the impedance transformation effect of the absorber can also help to explain the effectiveness of the quarter wavelength matching layer. When the impedance difference between the transmission line and the load is too large, the reflection of EMW is easily caused. In this case, according to the calculation formula of input impedance, by inserting a quarter wavelength odd multiple length transmission line, also called impedance converter, between the transmission line and the load, the gap between them can be further narrowed and the impedance matching can be approached. For the design process of the absorber, the absorber is equivalent to the inserted quarter wavelength impedance converter. When the absorber is inserted between the air and the load, the situation is shown in Fig. 3. The equation can be obtained as follows:

$$Z_{in} = Z_M \frac{Z_L + jZ_M \tan(\frac{2\pi}{\lambda}z)}{Z_M + jZ_L \tan(\frac{2\pi}{\lambda}z)} = \frac{Z_0^2}{Z_L} \Rightarrow Z_M = \sqrt{Z_L Z_0} \quad (4)$$

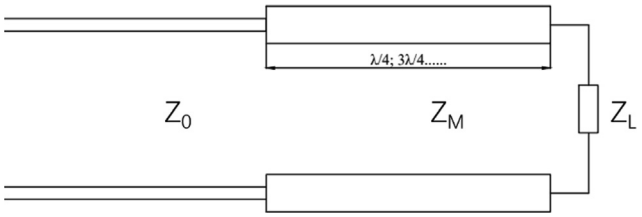


Fig. 3. Schematic diagram of impedance transformation.

Eq. (4) shows that the condition for no reflected wave at the interface between air and the absorber is that the characteristic impedance of the absorber is the geometric average of the characteristic impedance of the air and the load impedance, and the thickness of the absorber is an odd multiple of a quarter wavelength. In conclusion, the effectiveness of quarter wavelength matching layer can be explained by interference cancellation, multiple reflection and impedance transformation.

3. Conclusion

In conclusion, the calculation of impedance matching effect and RL must be based on the correct understanding of the transmission line model, so that the absorption capacity of the absorber to the EMW can be characterized. When the actual working environment of the absorber changes or the structure of the absorber changes, the corresponding transmission line model should be simplified first, and the calculation formula should be derived from the basis of the input impedance. The quarter wavelength matching layer can not only realize the interference cancellation of EMW, but also play a positive role in impedance matching through impedance transformation.

CRediT authorship contribution statement

Tao Wang: Writing - original draft, Visualization, Investigation. **Geng Chen:** Writing - review & editing. **Jiahao Zhu:** Validation, Methodology. **Hang Gong:** Writing - review & editing. **Limin Zhang:** Writing - review & editing, Funding acquisition. **Hongjing Wu:** Writing - review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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